

Commercial Analytics & Financial Modelling Capstone

Technical report | Tunji Ologun | Data Analyst | Synthetic data

1. Problem

NTech's Commercial team reports on procurement, fleet, workforce, agency and finance data held in separate systems. Names disagree, invoice IDs collide and the monthly figures are rebuilt by hand. The brief was to produce one trusted view of 2025 spend, a savings tracker the team can defend, and a 2026 model showing what pay, agency and sourcing decisions do to EBITDA, without handling personal data.

2. Data

Seven synthetic CSVs cover 20 sites in four regions: 1,800 procurement invoices, 1,159 fleet rows, 1,193 workforce rows, 492 agency rows and 240 site-month finance rows, plus site and category-target reference tables. Faults were left in deliberately: a missing supplier, site and amount, a reused invoice ID, 15 invoices with casing errors, and a supplier lookup holding 13 duplicate rows and one conflicting mapping.

3. Approach

Raw tables are never edited. Cleaned_Procurement standardises supplier and category with case-insensitive INDEX/MATCH against a rebuilt 24-row supplier master, adds site town with VLOOKUP, and classifies each row through a nested IF hierarchy. A true duplicate needs the same ID, supplier and amount, and only the second copy is excluded; a reused ID with different details stays in and is referred. Data_Quality counts and logs every exception, then reconciles raw to cleaned to the finance ledger with PASS/FAIL tests against an input tolerance.

Analysis sheets cover supplier concentration and contract compliance, site budget variance with RAG status, fleet cost per mile, overtime, absence and agency share of staff cost. The Dashboard has four drop-downs (region, category, month, scenario) feeding a calculation sheet built on wildcard and date-window SUMIFS, so 15 KPI cards, four self-writing insight sentences and eight charts move together. A LARGE/INDEX/MATCH block ranks the top ten suppliers for any filter combination.

4. Financial model

Each 2026 line is its 2025 actual moved by one driver: revenue growth, procurement savings, agency rate inflation, fleet efficiency, payroll inflation and other Opex inflation. Conservative, Base, Aggressive and a user-defined Custom case switch through CHOOSE/MATCH. The sheet adds an EBITDA bridge drawn as a waterfall, monthly phasing on 2025 seasonality, a forecast-versus-budget line and two formula sensitivity grids. Built-in checks confirm that the selected scenario, the bridge and the monthly total all tie back to the annual forecast.

5. Findings

Six of 1,800 invoices (0.3%) raised exceptions. The £5,662 gap between cleaned procurement and the finance ledger is exactly the invoice with no site code, which the ledger could not post to a cost centre.

82% of procurement spend (£11.5m) is off-contract. Agency Labour is the largest category at £3.19m, and its biggest supplier, NurseLink (£1.16m), has no contract. All three agencies charge identical rates by role, so agency savings will come from hours and grade mix rather than rate negotiation.

In the Base case 2026 EBITDA rises from £26.1m to £28.1m (31.5% margin); the Conservative case falls to £25.6m. One point of payroll inflation costs £381k, 2.7 times the value of a point of procurement saving, which makes the pay settlement the single biggest assumption in the plan. The savings tracker forecasts 81% of the £782k category target, with Agency Labour and Facilities off track.

6. Automation

Six VBA modules support the monthly cycle. Validation uses arrays and a Dictionary, logs exceptions to DQ_Log and leaves source text alone. Consolidation loops a folder, rejects files whose headers differ and stamps the source file

on every row. Refresh recalculates, refreshes the PivotTable and writes reconciliation status onto the Dashboard. Export produces a single PDF or one per region. A scenario runner logs all four cases, and a setup macro draws the Dashboard buttons. Power Query M and SQL versions of the cleansing and checks are included for the desirable skills.

7. GDPR and limitations

The data contains no names, identifiers or dates of birth; workforce figures are aggregated by site, month and role, and exported PDFs carry aggregates only. All figures are synthetic and illustrative. The model ignores volume growth, price inflation on the procurement base and everything below EBITDA. The macros target Excel desktop and should be run first on a copy.